Open-Source Continuous-Culture Bioreactors:

A Correction, a Comparison, a Two-Instrument Proposal, and a Path to AI-Directed Evolution

Morgan Hough • prepared with AI assistance (Claude, Anthropic)

The thesis: three purposes, one lifecycle

Low-cost open bioreactors are compared as if they compete for one role. They do not — the field serves **three purposes**, stages of a lifecycle, not a ranking:

proto **characterise** — hold a defined state, measure richly in situ. *Champion: Chi.Bio (an instrument).*

devo **evolve** — weeks of unattended selection producing *archived, replicated* evidence. *Champion: none yet.*

bior **scale out** — many reliable reactors, turnkey. *Champions: Pioreactor, eVOLVER.*

proto → **devo** → bior: prototype, evolve, produce. A machine optimised for one is measurably worse at another — so build purpose-built instruments and **share the parts**, don't merge.

Three lineages, seventy years

Each purpose is its own research tradition — a founding paper and a landmark demonstration:

	Founding lineage	**Landmark**
proto	chemostat as instrument + closed-loop control (Monod; Novick–Szilard 1950; Milias-Argeitis 2011)	optogenetic feedback holding expression + growth (2016)
devo	chemostat selection → LTEE → PACE (Novick–Szilard 1950; Lenski 1991; Esvelt 2011)	aerobic citrate use in the LTEE (2012)
bior	continuous culture for production → arrays → eVOLVER (Herbert 1956; Miller 2013; Wong 2018)	eVOLVER ePACE: compact Cas9 (2022)

Now one $300 device can, in principle, serve all three — and the argument here is that it should not try to at once.

Six platforms, like for like

	Chi.Bio	**OpenEvo**	**Pioreactor**	**eVOLVER**	**EVE**
Purpose	**proto**	**devo**	bior	bior	**devo**
Paper	**yes**	**yes**	none	**yes**	**yes**
Parallelism	**8**	1	**cluster**	**16**	3+
Rich sensing	**spectro.**	OD	plugin	config.	OD
Archiving	no	no	no	no	no
Cost / unit	$300	$300	$329	<$2k/16	**$115**
Build	PCB	**guided**	**buy**	shop	mod.

Two patterns: nobody archives (the field's blind spot), and the only paperless platform is the only pure product.

OpenEvo: a fair correction

The recent OpenEvo preprint is a real contribution — cheapest buildable selection cycler, best manual — but has correctable flaws:

- Error: abstract "55% faster" vs. results "36% increase" — same data, one mislabelled (7.6→4.9 h is +55% rate).
- Error: claims Pioreactor hardware is closed — it is **CC BY-SA 4.0** open.
- Method: $n = 1$, no replicates ⇒ no causal attribution; no archiving; default mapper, not `breseq`, for IS-mediated deletions.

Why `devo` has no champion

devo is defined by **evidence**, not actuation. OpenEvo actuates but produces no learnable signal. **The apparatus is the champion; a smarter vessel is not.** What the actuator must connect to (only one is new hardware):

- replicate manifold (borrow Pioreactor's cluster) → convergence
- **automated chilled archiver** — the field's missing freezer → fitness
- sequencing + `breseq` → correct variant calls
- fitness loop (reuses OD hardware); selection design (growth-coupling)

Three biopunk builds & what they buy

proto — updated Chi.Bio. Chi.Bio core + OpenEvo gas vessel + through-wall DO/pH optodes + C12880MA spectrometer + Pi. *Benefit:* the most complete non-contact in-situ characterisation instrument in open hardware.

devo — evolution + evidence. Pioreactor cluster (or OpenEvo base) + 3-media fluidics + replicate fleet + **archiver** + `breseq` + fitness assay. *Benefit:* fills the empty **devo** slot — first open platform with ***publishable*** evolution evidence.

bior — Pioreactor + add-ons. Buy the cluster; add the shared optode module (plugin), eVOLVER sleeves only where needed. *Benefit:* reliable parallel cultivation now — mostly bought, not built.

Three configurations of **one shared platform**: Raspberry Pi + one module footprint + one calibration reference.

The goal: AI-directed evolution

OpenEvo *pitched* "AI to supervise the evolution" but built only the actuator. **The observation layer an AI needs *is* the evidence apparatus** — so **apparatus = champion, AI = autopilot**; build the apparatus first, then close the loop:

- **observe** growth + fitness + genotype (`breseq`)
- **act** on media / light / dilution + **CRISPR-targeted mutation** (EvolvR, an **IGI** tool)
- **learn** rule-based → Bayesian opt → RL, across the fleet

CRISPR sharpens every layer: EvolvR (targeted diversification), CAMERA (in-cell recorder), Cas-coupled selection. *Prove it: AI-directed vs. fixed schedule, same target.*

Take-aways

1. Three purposes, not one contest — different winners.
2. Evolution has no champion: nobody archives or replicates. *That is the opening.*
3. The killer feature nobody has shipped: an **automated chilled archiver**.
4. Build the apparatus first; the AI autopilot follows.

github.com/m9h/biopunk-bioreactor